

G. OZAN BOZDAG

Senior Research Scientist | Adjunct Instructor

School of Biological Sciences, Georgia Institute of Technology

310 Ferst Drive NW, Atlanta, GA 30332

gonensin.bozdag@biology.gatech.edu | ozanbozdag.github.io | multee.org

EDUCATION

2016 Ph.D., Evolutionary Biology — Max-Planck Institute for Evolutionary Biology, Plön, Germany

2010 M.Sc., Molecular Biology and Genetics — Izmir Institute of Technology, Turkey

2006 B.Sc., Biochemistry — Ege University, Izmir, Turkey

ACADEMIC POSITIONS

2024–Present Senior Research Scientist, Georgia Institute of Technology

2022–Present Adjunct Instructor, Georgia Institute of Technology

2020–2024 Research Scientist II, Georgia Institute of Technology

2016–2020 Postdoctoral Fellow, Georgia Institute of Technology

2008–2011 Research & Teaching Assistant, Izmir Institute of Technology, Turkey

RESEARCH INTERESTS

Origins of multicellularity, speciation genetics, evolution of selfish genetic elements, evolution of genome architecture, yeast genetics, experimental evolution, omics/bioinformatics.

PUBLICATIONS

Peer-Reviewed Journal Articles & Preprints

1. Bozdag GO, Tong K, Yunker PJ, Herron MD, Ratcliff WC. Overcoming key challenges in the evolutionary transition to multicellularity. *Nature* (2026) doi.org/10.1038/s41586-026-11038-w
2. Yoon K, Yoon H, Tadesse K, Ling Z, Mandracchia B, Datta S, Bozdag GO, Burnetti AJ, Ratcliff WC, Jia S. Real-time, high-throughput super-resolution microscopy via panoramic integration. *Nature Communications* (2025) doi.org/10.1038/s41467-025-64368-0
3. Tong K, Datta S, et al., Bozdag GO & Ratcliff WC. Whole-genome duplication in the Multicellularity Long Term Evolution Experiment. *Nature* (2025) doi.org/10.1038/s41586-025-08689-6
4. Narayanasamy N, Bingham E, Fadero T, Bozdag GO, Ratcliff WC, Yunker P, Thutupalli S. Metabolically driven flows enable exponential growth in macroscopic multicellular yeast. *Science Advances* (2025) DOI: 10.1126/sciadv.adr6399
5. Pineau R, Kahn PC, Lac DT, Denning M, Wong W, Ratcliff WC, Bozdag GO. Experimental evolution of multicellularity via cuboidal packing in fission yeast. *Evolution Letters* (2024) doi.org/10.1093/evlett/qrae024
6. Bozdag GO, Szeinbaum N, Conlin PL, Chen K, Mestre Fos S, Garcia A, Penev PI, Schaible GA, Trubl G. Major biological innovations in the history of life on Earth. *Astrobiology* (2024) doi.org/10.1089/ast.2021.0119
7. Pineau R, et al., Bozdag GO & Ratcliff WC. Emergence and maintenance of stable coexistence during a long-term multicellular evolution experiment. *Nature Ecology & Evolution* (2024) doi.org/10.1038/s41559-024-02367-y
8. Montrose K, Lac DT, Burnetti AJ, Tong K, Bozdag GO, Hukkanen M, Ratcliff WC, Saarikangas J. Proteostatic tuning underpins the evolution of novel multicellular traits. *Science Advances* (2024) DOI: 10.1126/sciadv.adn2706
9. Day TC, Zamani-Dahaj SA, Bozdag GO, et al. Morphological entanglement in living systems. *Physical Review X* (2024) doi.org/10.1103/PhysRevX.14.011008

10. Bozdag GO, Zamani-Dahaj SA, Day TC, et al., Ratcliff WC. *De novo* evolution of macroscopic multicellularity. *Nature* (2023) doi.org/10.1038/s41586-023-06052-1
11. Bozdag GO and Ono J. Evolution and molecular basis of reproductive isolation. *Current Opinion in Genetics & Development* (2022) doi.org/10.1016/j.gde.2022.101952
12. Tong K, Bozdag GO, Ratcliff WC. Selective drivers of simple multicellularity. *Current Opinion in Microbiology* (2022) doi.org/10.1016/j.mib.2022.102141
13. Bozdag GO, Libby E, Pineau R, Reinhard C, Ratcliff WC. Oxygen suppression of macroscopic multicellularity. *Nature Communications* (2021) doi.org/10.1038/s41467-021-23104-0
14. Bozdag GO, Ono J, Denton J, Karakoc E, Hunter N, Leu JY, and Greig D. Breaking a species barrier by enabling hybrid recombination. *Current Biology* (2021) doi.org/10.1016/j.cub.2020.12.038
15. Pentz JT, Márquez-Zacarias P, Bozdag GO, Burnetti A, Yunker PJ, Libby E, & Ratcliff WC. Ecological advantages and evolutionary limitations of aggregative multicellular development. *Current Biology* (2020) doi: 10.1016/j.cub.2020.08.006
16. Bozdag GO, Ono J, Denton J, Karakoc E, Hunter N, Leu JY, and Greig D. Engineering recombination between diverged yeast species reveals speciation genes. *bioRxiv* (Preprint, 2019) doi.org/10.1101/755165
17. Bozdag GO & Greig D. The genetics of a putative social trait in natural populations of yeast. *Molecular Ecology* (2014) 23:5061–5071 <https://doi.org/10.1111/mec.12904>
18. Telli M, Kulkoyluoglu O, Bozdag GO, Yavuzatmaca M. Comparative phylogenetic analyses of cave- and surface-water Ostracoda based on mitochondrial CO-I. *Cave and Karst Science* (2016) 43(2):65–74
19. Bozdag GO, Kaya A, Koc A, Noll GA, Prüfer D, Karakaya HC. Characterization of a cDNA from *Beta maritima* that confers nickel tolerance in yeast. *Gene* (2014) 538(2):251–7 DOI: 10.1016/j.gene.2014.01.052
20. Erbasol I, Bozdag GO, Koc A, Pedas P, Karakaya HC. Characterization of two genes encoding metal tolerance proteins from *Beta vulgaris* subspecies *maritima*. *Biometals* (2013) 26(5):795–804 DOI: 10.1007/s10534-013-9658-7
21. Atik AE, Bozdag GO, Akinci E, Kaya A, Koç A, Yalcin T, Karakaya HC. Proteomic changes during boron tolerance in barley and the role of vacuolar proton-translocating ATPase subunit E. *Turkish Journal of Botany* (2011) 35(4):379–388 doi:10.3906/bot-1007-29
22. Bozdag GO, Uluisik I, Gulculer GS, Karakaya HC, Koc A. Roles of ATR1 paralogs YMR279c and YOR378w in boron stress tolerance. *Biochemical and Biophysical Research Communications* (2011) 409(4):748–51 DOI: 10.1016/j.bbrc.2011.05.080

Grants & Awards

- 2025–2030 NSF LTREB Grant (PI), \$596,940 — Georgia Institute of Technology
- 2025 CIO Honor Roll: Student Recognition of Excellence in Teaching, Microbiology Lab — Georgia Tech
- 2024 CIO Honor Roll: Student Recognition of Excellence in Teaching, Evolutionary Biology — Georgia Tech
- 2022 Center for Microbial Dynamics and Infection Prize for Postdoctoral Research — Georgia Tech
- 2019 Best Talk, Evolution of Complex Life Conference, USA
- 2015 Best Oral Presentation, EEBST Symposium, Turkey
- 2014 Best Talk, EEBST Symposium, Turkey
- 2012–2016 Ph.D. Fellowship, Max-Planck Research School for Evolutionary Biology, Germany

Invited Talks & Conference Presentations

- 2026 Astrobiology Science Conference — "From Microscopic to Macroscopic Size: Long-Term Experimental Evolution of Multicellularity in Yeast"
- 2025 Oregon State University, Dept. of Integrative Biology Seminar — "Replaying the Evolution of Multicellularity in a Test Tube"
- 2025 Kennesaw State University Seminar — "Replaying the Evolution of Multicellularity in a Test Tube"

2025 Darwin Day Talks, Georgia Tech — "Predictability of Evolution: Lessons from a Multicellularity Evolution Experiment"

2024 Suddath Symposium, Georgia Tech — "A Thousand Days of Change: Snowflake Yeast's Path to Multicellularity"

2022 Evolution Meeting, Cleveland — "Experimental Evolution of Macroscopic Multicellularity"

2021 Aykut Kence Evolution Conference, METU — "Long-Term Evolution of Multicellularity"

2021 Gregor Mendel Institute, Vienna Biocenter — "De novo Evolution of Macroscopic Multicellularity"

2019 Gordon Research Seminar, New Hampshire — "Oxygen and the Experimental Evolution of Macroscopic Multicellular Size"

2018 Yeast Genetics Meeting, Stanford — "Oxygen and the Evolution of Multicellular Size"

Selected Press Coverage

Astrobiology (2025): Whole-genome Duplication Drives Long-Term Adaptation

The New York Times (2023): An Experiment Repeated 600 Times Finds Hints to Evolution's Secrets

The Atlantic (2023): One of Evolution's Biggest Moments Was Re-Created in a Year

National Geographic (2021): Evolving Globes of Yeast May Unlock Mysteries of Multicellular Life

Quanta Magazine (2021): Single Cells Evolve Large Multicellular Forms in Just Two Years

Science Daily (2023, 2021): Multiple features on multicellularity research

TEACHING

Instructor of Record — Full-Semester Courses

Semester	Course	Enrollment	Eval. Score
Spring 2026	Evolutionary Biology (BIOS 3600/BIOL 6600)	84	4.7/5
Fall 2025	Microbiology Lab (BIOS 3381)	44	4.95/5
Fall 2024	Microbiology Lab (BIOS 3381)	48	4.8/5
Spring 2024	Evolutionary Biology (BIOS 3600/BIOL 6600)	66	4.85/5
Fall 2023	Evolutionary Biology (BIOS 3600/BIOL 6600)	70	4.7/5
Spring 2023	Evolutionary Biology (BIOS 3600/BIOL 6600)	98	4.7/5
Fall 2022	Communicating Biological Research (BIOS 4460)	9	4.6/5

Guest Lectures

Seminal Papers in Astrobiology (EAS/CHEM/BIOS 8802) — Evolution of Complex Life (1 lecture, 2026)

Genetics (BIOS 2600) — Population Genetics (1 lecture, 2024)

Evolution of Complex Life (BIOS 2550/8803) — Origin of Eukaryotic Cell (2 lectures; 2018, 2019, 2022, 2024)

Evolutionary Biology (BIOS 3600/BIOL 6600) — Experimental Evolution, Speciation (3 guest lectures; 2017, 2018, 2019)

Workshops

2021 Evolutionary Genomics Winter School (Virtual) — Introduction to Genome Data Analysis Tools

2019 Evolutionary Genomics Winter School — Introduction to Genome Data Analysis Tools

2018 Evolutionary Genomics Winter School, Hacettepe University — Linux Bash for Genomics & Transcriptomics

Teaching Assistantships — Izmir Institute of Technology (M.Sc., 2008–2010)

Molecular Biology Laboratory (MBG 212) — Full semesters, 2008–2010

Biochemistry Laboratory (MBG 311) — Full semesters, 2009–2010

Genetics Laboratory — Full semester, 2010

Current Topics in Molecular Biology Laboratory — 4-week module

MENTORING & ADVISING

Ph.D. Committee Membership

Hikaru Katani, Biological Sciences (2024–present) (Co-advisor)

Luis Felipe Pérez, Quantitative Biosciences Program (2023–present) (Co-advisor)

Maryam Heiri, Quantitative Biosciences Program (2023–present)

Sayantan Datta, Quantitative Biosciences Program (2022–present) (Co-advisor)

Autumn Peterson, Quantitative Biosciences Program (2020–2026)

Raymond R. Copeland, Quantitative Biosciences Program (2020–2025)

Rozenn Pineau, Quantitative Biosciences Program (2018–2023) (Co-advisor)

Faculty Reader / Advisor for Undergraduate Senior Research (BIOS 4699)

Gillian T. Myers (2025, Reader & Advisor); Li Ying (2024, Reader); Mia Denning (2024, Reader & Advisor);
Perna Kotil (2023, Reader); Vivian Cheng (2021, Reader & Advisor); Daniella Haas (2021, Reader)

Letters of Recommendation

Authored recommendation letters for 30+ students at all levels (undergraduate, graduate, postdoctoral) for graduate programs, medical/dental/PA school, fellowships (including NSF GRFP), and research positions, supporting their transitions to external institutions.

SERVICE

University Service

2025–Present Academic Research Senate Member, Georgia Institute of Technology

2025, 2026 School of Biological Sciences Promotion Committee, Georgia Institute of Technology

Conference Organization

2026 Diversification of Life Meeting, Atlanta — Multicellularity Session Chair

2026 Evolution Conference, Cleveland — W.D. Hamilton Award & ASN Ruth Patrick Award Judge

2024 ASM Microbe, Atlanta — Microbial Experimental Evolution Session Co-convenor

2021 Ecology and Evolutionary Biology Symposium, Istanbul — Session Chair

2015 Max-Planck Evolutionary Biology Aquavit Symposium, Plön — Co-organizer

2015 Quantitative Evolutionary Biology Workshop, Sirince — Co-organizer

Journal Peer Review

Science Advances, Nature Communications, Genome Research, eLife, Molecular Biology and Evolution, Genetics, Communications Biology, Journal of Evolutionary Biology, Journal of Molecular Evolution, Plant Communications (Cell Press), BioEssays

Grant Proposal Review

2025–2026 PURA Salary Award Reviews (8 proposals per term), Georgia Tech

2025 Alexander von Humboldt Foundation Research Fellowship Programme, Germany

Professional Memberships

Ecology and Evolutionary Biology Society